Module 2 Group Activity

Matrix Theory

1. For each of the following, determine whether U is a subspace of V. You may assume the typical addition and scalar multiplication for V. Make sure to justify your answers!

   (a) $V = \mathbb{C}^3$ as a complex vector space, $U = \mathbb{C}^2$

   (b) $V = \mathbb{R}^{2\times 2}$ as a real vector space, $U = \{A \in \mathbb{R}^{2\times 2} : A^2 = 0\}$

   (c) $V = \mathcal{P}_2(\mathbb{C})$ as a complex vector space, $U = \{a + bt + ct^2 : a, b, c \in \mathbb{C}, a = -2c\}$

   (d) $V = \mathcal{C}([0,1], \mathbb{R})$ as a real vector space, $U = \{f \in V : f(1/2) \geq 0\}$.
   Note. U is the set of functions $f \in V$ such that f evaluated at $1/2$ is positive. For example, consider $f_1(t) = t + 1$ and $f_2(t) = t^2 - 1$. $f_1, f_2 \in V$ as both are continuous, real-valued functions on $[0, 1]$. However, $f_1 \in U$ since $f_1(1/2) = 3/2 \geq 0$ while $f_2 \notin U$ since $f_2(1/2) = -3/4 < 0$.